

Demonstrating Electrical Head Actuation: Enabling Interactive Systems to Directly Manipulate Head Orientation

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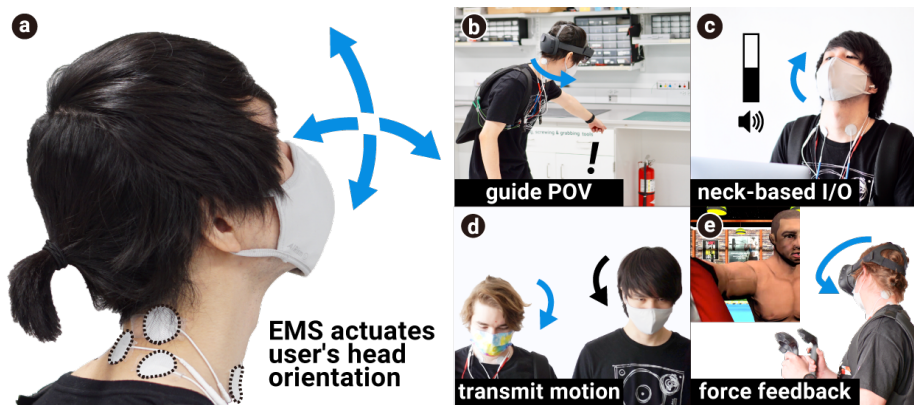


Figure 1: (a) We demonstrate a novel interface concept that enables interactive systems to directly actuate the user's head orientation around its yaw (i.e., left/right) and pitch (i.e., up/down) axis via electrical muscle stimulation (EMS) to the neck muscles. Actuating the neck muscles with EMS is uncharted territory for interactive systems, yet, enabling it opens up a range of applications not possible before: (b) directly changing the user's point-of-view to locate target objects in a fire safety training; (c) a sound controller that uses neck movements as both input and output; (d) transmitting one users' neck movement to another; and, (e) rendering force feedback from VR punches on the head.

ABSTRACT

We demonstrate a novel interface concept in which interactive systems directly manipulate the user's head orientation. We implement this using electrical-muscle-stimulation (EMS) of the neck muscles, which turns the head around its yaw (left/right) and pitch (up/down) axis. As the first exploration of EMS for head actuation, we characterized which muscles can be robustly actuated. Then, we demonstrated how it enables interactions not possible before by building a range of applications, such as (1) directly changing the user's head orientation to locate objects in AR; (2) a sound controller that uses neck movements as both input and output; (3) synchronizing head orientations of two users, which enables a user to communicate head nods to another user while listening to music; and (4) rendering force feedback from VR punches on the head by actuating the user's neck.

CCS CONCEPTS

• **Hardware** → Emerging technologies.

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Electrical Muscle Stimulation, Haptics, Augmented Reality, Virtual Reality

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1 INTRODUCTION

While head movements are integral in life and, in many interactive systems (e.g., scanning one's surroundings in VR/AR or as non-verbal cues in telepresence), researchers have rarely explored ways to directly utilize head movements as output. When it comes to haptic actuation, researchers predominately focus on actuating the user's hands [15] arms [10], or feet [12]. Even though actuating the limb's extremities has proven promising for guiding user's touch [1], movements [9] and even walking direction [12], very few interactive systems explored the potential of *directly actuating head movement*.

There have been four types of explorations in actuating the head: (1) emergent exoskeletons used only in medical rehabilitation [17]; (2) mounting a flywheel to the VR headset to render inertial forces, which uses the gyroscopic effect as the user's head is actively

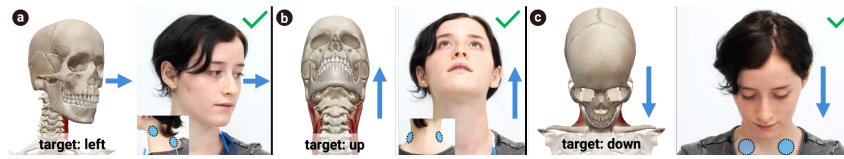


Figure 2: We found electrode placements that robustly turn the head left/right (a); nod the head up (b) and down (c) with EMS.

moving [2]; (3) mechanically applying pressure to the head causing it to jerk in one direction, denoted as hanger reflex [5]; and (4) using air jets mounted to the VR headset to render accelerations of the head [6]. Unfortunately, while all the above haptic interfaces are promising, they are built around the assumption that one can just attach actuators to a headset or, even more dramatically, that all users will be excited to attach actuators on their face, forehead, etc. Thus, these are mostly limited to VR, where a headset is already present.

Instead, in this paper, we demonstrate a novel approach that empowers interactive systems with direct control of the user's head orientation by actuating the user's neck muscles with electrical muscle stimulation (EMS). Our system is depicted in Figure 1 (a). The fact that it builds on EMS results in a haptic device capable of actuating the user's head without requiring any attachments on the user's face/head or above the neck muscles—e.g., a simple turtleneck would cover our system's EMS electrodes entirely.

As the first investigation of head actuation via EMS, we first explored and discovered the electrode placement that robustly actuates the user's head orientation around its yaw (left/right) and pitch (up/down) axis (Figure 2). We achieved this by placing electrodes at all skin locations atop the nine superficial neck muscles [13] and observing the resulting movement while applying EMS (more details can be found in [16]). Now, we showcase how our electrical head actuation enables interactions not possible before by building a range of applications, such as directly changing the user's point-of-view to locate targets in a mixed-reality fire safety training and transmitting head nods from one user to another.

2 DEMONSTRATING HOW DIRECT HEAD ACTUATION ENABLES NEW APPLICATIONS

We implemented four classes of applications: (1) snapping to spatial information by guiding POV; (2) neck-based I/O interface; (3) transmitting neck movement to another user; and (4) force feedback around the head for immersive experiences. Although we exhibit each application in either mixed reality, virtual reality, or the real world, these applications can be implemented across all of them.

2.1 Guiding One's Point-of-view

Electrical head actuation can change the user's point-of-view (i.e., what they are looking at) by directly actuating their head orientation. We demonstrate this in the example of mixed reality (MR) application for fire safety training. Here, the user wears our system as well as a HoloLens 2 MR headset, which displays the virtual content and tracks their head. The user is learning about fire safety from a first-person perspective, instead of with training videos. As



Figure 3: (a) The user sees a fire; but, (b) cannot locate the fire extinguisher. (c) Our device actuates their POV to look at it.

depicted in Figure 3 (a), they see a fire breaking out at the solder station through the MR headset. Their goal is to put out the fire. The MR interface visually depicts the next instruction to the user, “Put out the fire”. However, the user is having difficulties locating the fire extinguisher, which we depict in Figure 3 (b). Instead of the GUI elements that MR typically make use of to indicate off-screen targets, our system relies on a novel approach: it actuates the user's head to the lower-left, which is depicted in Figure 3 (c) so that the fire extinguisher comes into their view—in other words, our system actuates the head, which in turn guides the user's point-of-view (POV).

2.2 Neck-based I/O Interface for Controlling Sound Volume

In this application, we depict how we built a type of application, in which a UI uses head movements not just as input but also as output, which demonstrates how our technique can be employed to close an I/O loop in the neck. Here, we demonstrate this with the example of a simple music volume controller that the user adjusts using head movements. As depicted in Figure 4 (a), the user activates the system by using a trigger gesture (a triangular head motion detected by the user's AirPods Pro earbuds with IMU tracking), while listening to music; (b) our system immediately takes over and renders the current value of the music volume by moving the user's head up with EMS; (c,d) now, by moving their head voluntarily up/down, the user adjusts the volume, and as they hit the ends of the slider scale (i.e., maximum/minimum volume), the EMS pulls their head back in the corresponding direction to notify this; and, (e) when finished adjusting it, the user turns their head right to indicate the interaction is finished. Finally, while this example is a type of symmetric proprioceptive interaction (as in [8]), it is the first to extend this to the neck, which enables hands-free interaction.



Figure 4: Our music volume controller depicts how a UI can use neck movements for both input and output.

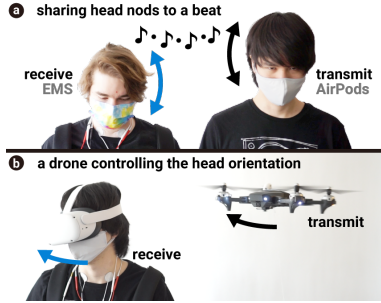


Figure 5: Our device can be used to transmit the head orientation between two users. (a) Two users nod to a beat in sync. (b) A drone’s rotation dictates the user’s neck movement.

2.3 Transmitting Neck Movement to Another User

We leverage our head actuation to transmit the movement of one user to another’s neck. In Figure 5, we depict a simple application that allows a user to nod their head to a beat and transmit that rhythmic neck movement to a remote user wearing our system (Figure 5a). This is inspired by *bioSync* [11] but allows sharing neck movement rather than that of the arms. By attaching a tracker to a remote-controlled drone, it is even possible to transmit the drone’s rotational movement to the user’s neck (Figure 5b); this allows a new type of human-drone interaction by inverting the concept shown in *Flying Head* [3].

2.4 Rendering the Force Feedback of Punches in VR Boxing

While EMS has been explored in VR for creating force feedback, all existing systems mostly only provide forces to the user’s arm, forearm, wrist, etc. Instead, in Figure 6, we demonstrate a boxing simulator in which the user feels the impact of left/right punches and uppercuts. This is inspired by *Impacto* [7] but depicts a novel use of EMS in that it allows the user to feel directional force feedback on *their head*.

3 IMPLEMENTATION

We now provide the necessary technical details. Additionally, all source code and materials are made publicly available¹.

3.1 Hardware

Figure 7 depicts our wearable hardware setup that ensures a user’s mobility. The system stimulates the user’s neck muscles at 30 Hz

¹<https://lab.plopes.org/#electrical-head-actuation>



Figure 6: (a) The user is experiencing VR boxing. Our system renders force feedback of the (b, c) left/right hooks and (d) uppercut.

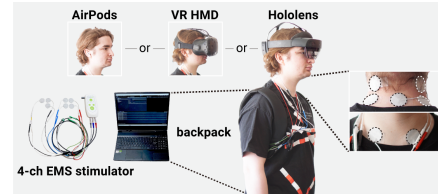


Figure 7: Our wearable system setup.

using a medical-grade electrical muscle stimulator (HASOMED RehaMove3) which the user carries in a slim backpack. Our system utilizes four channels of the EMS stimulator to actuate the head accordingly to the electrode placement found in our preliminary exploration: (1) turn left by stimulating the left side of the *splenius capitis*; (2) turn right by stimulating the right side of the *splenius capitis*; (3) nod up by stimulating the *splenius cervicis*; and (4) nod down by stimulating the *sternocleidomastoid*.

For tracking head orientation, we made our system compatible with HoloLens 2, HTC VIVE, and AirPods Pro earbuds, such that the user equips one of them based on the application type (i.e., MR, VR, and real environment). The interface between the EMS stimulator and the tracking device(s) is a laptop (MSI GL65 Leopard) running our Unity3D applications; the laptop is also stored in the user’s backpack.

3.2 EMS Control Loop

To actuate the user’s head to a target orientation, we implemented a PID controller, designed specifically for neck EMS. The PID controller regulates the pulse-width of the EMS impulses between the range of 0 μ s and 300 μ s. In each cycle, the system calculates the error (e_x , e_y) between the user’s head orientation, given by the current orientation (obtained from tracking), and the target orientation.

Note that the x axis here represents head rotations around the yaw axis (i.e., turn left/right) and the y axis represents head rotations around the pitch axis (i.e., nod up/down). With the input of e_x and e_y , our PID controller individually computes PW_x and PW_y , which are final EMS pulse-widths for actuating the user's neck to turn horizontally and/or nod vertically towards the target respectively. Our controller is dual in that, if PW_x is positive, it will be set as the pulse-width for the right-channel; if PW_x is negative, it will be negated and set as the pulse-width for the left-channel; similarly, if PW_y is positive, it will be set as the pulse-width for the up-channel, and vice-versa for the down-channel. For instance, in one cycle, $(PW_x, PW_y) = (-100, 200)$ dictates that left-channel outputs a 100 μ s pulse and up-channel outputs a 200 μ s pulse (i.e., no pulses are output from right-channel and down-channel). Via initial pilots, we settled on the following PID coefficients (Kp, Ki, Kd): (16, 4, 2) for static target acquisition. Finally, note that our PID controller works without any voluntary movement by users; if the neck's own springiness or the user moves away from the target, the PID will push it back without user intervention.

3.3 Tracking, Display, Communication and Latency

Our head actuation system was implemented to be agnostic of the tracking system as long as it provides rotational tracking of the user's head. We implemented bridges between our system and the three external tracking systems, which afford different use cases. Note that in all our applications, users tend to wear the complete system on their body, typically in a backpack. Naturally, the choice of a tracking system impacts the quality of the resulting applications, i.e., using room-level positional tracking (e.g., HTC VIVE or Hololens 2, which we use for our VR and MR applications) allows applications to actuate the user's head to point to targets in physical space; conversely, using only inertial-based tracking systems (not positional) allows applications that actuate the user's head with respect to the inertial frame of reference (e.g., IMUs from AirPods Pro, which we use for our *neck-based I/O or transmitting neck movement* applications).

MR tracking. We leverage the tracking of the *Hololens 2* to track both the user's head position in space and its rotation. This data is streamed in real-time from the headset to the laptop running the Unity application via Wi-Fi, built on top of the *Holographic Remoting Player API*. Using the same communication channel, the laptop streams and displays visual MR scenes to the user's headset.

VR tracking. We leverage the tracking from the *HTC Vive* system, using its base stations and the headset.

Mobile tracking. In our most mobile and minimal tracking implementation, we leverage *AirPods Pro*, which tracks only the user's rotation with its inertial measurement unit. The sensor data is retrieved via Apple *Core Motion API*. Since the API can only be accessed from iOS, the sensor data is first streamed from an iOS device (*iPhone 11*) and relayed to the laptop via Open Sound Control over Wi-Fi.

End-to-end latency. In our most extreme applications that run over the Wi-Fi (e.g., transmitting head nods to a remote user) our system has an end-to-end latency of around 246 ms (calculated from high-speed video).

4 CONCLUSIONS

We demonstrated a novel interface concept in which the interactive system actuates the user's head orientation. We implemented this concept by applying electrical muscle stimulation (EMS) to the user's neck muscles, enabling our system to actuate the user's head orientation around its yaw and pitch axis. We believe our work will inspire researchers to further delve into this avenue of directly controlling the user's head orientation. We believe there are several ways that researchers can further expand on our work, and, to accelerate this, we have provided all the source code¹. Possible future investigations include exploring how electrical head actuation (1) interacts with eye-tracking and visual gaze redirection, (2) impacts the sense of agency [4], (3) might enable to control the user's neck posture over time (e.g., which has been explored also via a robotically actuated computer monitor [14]). Our work provides a foundation for researchers to explore these challenges and expand on our findings or system in novel directions.

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